

How can outcomes theory and DoView Planning be improved by mathematization?

Because it is a new theory, outcomes theory has not yet been extensively mathematized. Mathematization of a theory consists of using mathematical frameworks to systematically describe and explore it. Doing this can have a number of benefits. It can help you examine the internal consistency of the theory's principles. It also means that you may be able to extract from the existing theory's principles a smaller set of core axioms underlying it. Doing this can then mean that you can then identify new principles that logically follow from the core axioms, because you have captured the essence of the theory in those underlying axioms.

Another benefit of mathematizing a theory is that it can provide practical tools that can be used in applied versions of the theory. One example is that it would be good to be able to identify the best mathematical way of assessing the degree to which the logic within a particular DoView diagram is similar to the logic underlying a different DoView.

One possibly promising mathematical theory that could play a significant role in outcomes theory is category theory. Category theory provides a mathematical framework for understanding and representing relationships and processes using diagrams; this would seem to make it an ideal candidate for possible application to DoView Planning and outcomes theory.

Once you have mathematized aspects of DoView Planning, this may make it possible to automate checking, comparison, and optimization of outcome pathways using software tools. It could also help with modeling the effects of different interventions. Mathematizing DoView Planning could also help integrate it with the use of other mathematical tools that are related to planning and implementation, such as probability theory, logic, systems theory, and mathematical aspects of operations research.